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Course Name: design and analysis of Algorithm.

Q77

Selection Sort - Dataset with two
smallest equal values.
Aim:

to sort the dataset $[24, 7, 31, 7, 45, 12]$ using the selection (bubble force) algorithm and analyze its performance.

Problem Interpretation:

Selection Sort repeatedly finds the smallest element from the unsorted portion of the array and places it at the beginning.

Algorithm:

1. Start from the first element
2. Find the smallest element in the remaining unsorted array
3. Swap it with the current element
4. Repeat until the array is sorted.

Initial Array: $[24, 7, 31, 7, 45, 12]$

Pass-1

→ minimum = 7 (first occurrence)

→ Swap 24 and 7

→ Array: $[7, 24, 31, 7, 45, 12]$

Pass-2)

→ minimum = 7

→ swap 24 and 7

→ Array : [7, 7, 31, 24, 45, 12]

Pass-3)

→ minimum = 12

→ swap 31 and 12

→ Array : [7, 7, 12, 24, 45, 31]

Pass-4)

→ minimum = 24

→ No swap needed

→ Array : [7, 7, 12, 24, 45, 31]

Pass-5)

→ minimum = 31

→ swap 45 and 31

→ Array : [7, 7, 12, 24, 31, 45]

Final sorted Array

[7, 7, 12, 24, 31, 45]

comparisons

For 6 elements

→ Pass 1 = 5

→ Pass 2 = 4

→ Pass 3 = 3

→ Pass 4 = 2

→ Pass 5 = 1

swaps

* Pass 1 = 1

* Pass 2 = 1

* Pass 3 = 3 = 1

* Pass 4 = 0

* Pass 5 = 1

∴ Total comparisons

$$= 5 + 4 + 3 + 2 + 1 = 15$$

Total swaps = 4
Time complexity
→ Best case: $O(n)$
→ Average case: $O(n^2)$
→ Worst case: $O(n^2)$
Correctness verification

the final sorted array is

[7, 7, 12, 24, 31, 45]

All elements are in ascending order, so the algorithm works correctly.

Behaviour on equal minimum values.
→ the array contains two equal minimum values (7 and 7)

Results

→ The ^{data} set was successfully sorted using selection sort with 15 comparisons and 4 swaps, producing the final sorted array [7, 7, 12, 24, 31, 45].